# Students: Remember This

- When printing from a pdf from Adobe, you must select the option **"Page scaling: None."** Otherwise, Adobe will be "helpful" and mess up the margins.
- Remove this page but leave the next page for the proofreaders.
- Remove this page and the next one from the final version that is submitted for binding.
- Format checkers will mark English mistakes if they happen to see them, but they are not proofreaders. So passing the format check does not mean your spelling and grammar are correct, and mistakes that were not caught in one format check may be caught in another.

Additionally there are a few known issues with this template that you should check if you want to pass on your first try:

- Margin violations, called "overfull" or "underfull" `\hbox` or `\vbox`. (These are alerted to you in the log file or somewhere in the GUI of your editor.)
- Widows and orphans—single lines that are split from their paragraph at a page break.
- Spacing below `[h]` (inline) floats.
- Spacing between captions and their figures.
- Each figure must be referred to by an in-text reference before it is shown. *Even if it is on the same page as the figure*, it will be sent back to you.

Good advice for feedback from proofreaders in general: When the format for a certain type of entry, say a section heading, is marked as "to be corrected" (long section headings are supposed to be in inverted pyramid form), it is best to check the format of *all* these entries, as the format needs to be consistent overall.

## Notes for Proofreading and Format Checking

We appreciate the services provided by proofreaders and format checkers to assure uniformly high quality of documents produced at Louisiana Tech University. Because of the differences between mathematical documents and other documents, we request that the following be kept in mind.

- This document was typeset using Louisiana Tech's approved LaTeX template. Therefore most formatting should be a non-issue.
- Issues that should not require correction:
  - Global margins, order of sections, page numbering, title page, format of headings, table of contents, list of figures, list of tables, bibliography. (All approved when the template was created.)
  - LaTeX is *the* professional standard for mathematical typesetting. Equations are typeset with the LaTeX default options, which should not be adjusted.
  - Some built in font sizes cannot be changed. Font sizes for headings, etc., were approved, even if they may look a little different than in Microsoft Word documents.
- Some situations in which LaTeX' automatic formatting is less than optimal:
  - Margin infractions on individual lines. The global margins have been approved, but if the program does not know how to split a long term, it can spill into the margin. This is especially likely in typeset equations and in the lists of tables and figures and can and should be fixed.
  - Widows and orphans. Linebreaking is automatic and sometimes leaves the first line of a paragraph on the preceding page or puts the last line on the next page.
  - English spelling, grammar, punctuation, etc.
  - Gross infractions on the spacing around figures.
  - Placement of figures and tables relative to in-text call-outs.

# THESIS TITLE GOES HERE IN CAPITAL LETTERS, SPLIT FOR INVERTED PYRAMID FORM

by

Your Name, Previous degree(s)

A Dissertation/Thesis Presented in Partial Fulfillment
of the Requirements for the Degree
PhD/Masters of ...

COLLEGE OF ENGINEERING AND SCIENCE
LOUISIANA TECH UNIVERSITY

August 2025

Replace this page with the Signature Page.

## ABSTRACT

**Put your abstract here.** This LaTeX template for MS and Ph.D. theses automatically takes care of most formatting requirements for theses submitted at Louisiana Tech University. A short tutorial on LaTeX as well as an indication of issues that must be handled on a case-by-case basis is included.

If the abstract splits over two pages, LaTeX will take care of the page numbering.

Replace this page with the approval for scholarly dissemination form.

# DEDICATION

Place dedication here, if a dedication is desired. Otherwise, comment out the line `\include{tex/dedication}` in `thesis.tex`.

# TABLE OF CONTENTS

# LIST OF TABLES



So the entries above would need to be fixed by rewording to eliminate the spills. (And text like is not permissible.) Also note that to shrink the gap between the dotted line and the end of the entry, you should put the closing parenthesis `}` of the caption command right after the last word of the caption, not on the next line. Compare the entries for Table 3.1 (bad) and the rest of the entries here.

Also, you shouldn't forget to delete this note from `thesis.tex`.

# LIST OF FIGURES

# ACKNOWLEDGMENTS

Place acknowledgements here, if acknowledgements are desired. For an example, see below.

I (B. Schröder) thank Dr. Lisa Kuhn for letting me use her dissertation's source code as the basis for this template. The hard work was done by her. I only inserted hints and a tutorial on LaTeX that I had on hand.

If no acknowledgements are desired, comment out the line `\include{tex/acknowledgements}` in `thesis.tex`.

*Note that the above is an example of a margin violation that would need to be fixed manually. LaTeX usually does a good job splitting words correctly between two lines. Issues only arise with words from other languages or with math expressions, verbatim text, and other combinations which normally don't allow line breaks.*

# PREFACE

Place preface here, if a preface is desired. (Many CAM and MS theses do not have a preface, because the introduction serves the purpose that a preface would in a book.)

Otherwise, comment out the line `\include{tex/preface}` in `thesis.tex`.

# CHAPTER 1

# INTRODUCTION

**Put your introduction here.**

Typesetting in LaTeX is *the* way to communicate mathematics (journal articles, books, MS theses, doctoral dissertations) and is also popular in other disciplines, such as physics or computer science. This template should make the formatting of your thesis easier. It is set up so that, unless there are some really wide equations or figures, all margins should automatically be correct. Similarly, all parts are in the order required by the graduate school and the required tables (listing contents, figures, tables) are created automatically in the required format. The template is compiled by running LaTeX on the file `thesis.tex`. The only parts of `thesis.tex` that you should change are:

- `\include{...}` commands: add more to accommodate all the parts of your thesis and delete those that do not apply.
- `\usepackage{...}` commands: to import commands and environments that other people have created.
- Your own `\newcommand` statements.
- A few other setup commands that you may need to change (these are clearly labeled as needing your input).

Note that the text within the `itemize` environment here is indented by one tab. This is not necessary, but it helps give your document clearer structure. Similarly, this template uses a single line per sentence.

Although LaTeX takes care of all formatting automatically, its solutions aren't always ideal. Don't worry too much about any such mistakes until it's time to submit your thesis, at which point you can fix them manually. Thankfully, LaTeX can inform you of most typsetting errors it cannot correct. Overleaf and most LaTeX editors will show you any errors or warnings, but if you run LaTeX from the command line, then you should run `texlogfilter` on the resulting log file to see them. If you are going to use the command line to run LaTeX, you will find a makefile with all the necessary commands for building `thesis.pdf`. By the way, LaTeX may need to be run multiple times on a file to resolve all references; `latexmk` can automatically recompile your document as needed and even call other support programs as needed.

LaTeX has a variety of distributions for various platforms. These will manage all the packages, TeX engines, and auxiliary programs for you.

- MikTeX [1] is the standard LaTeX distribution for Windows and WinEdt [2] is a standard front end. WinEdt asks for a registration fee, but TeXnicCenter [3] is free.
- MacTeX [4] is the standard MacOS distribution (it's an idiot-proof version of TeX Live). It comes with the TeXShop editor.
- TeX Live [5] is the standard Linux distribution.
- Overleaf [6] is a convenient web app for hosting and editing your TeX projects.

Running using some of these distributions on your computer involves quite large download sizes (several gigabytes), but this means that you can run any package you want with full offline access to documentation (run `texdoc <package>` from the command line with a TEX Live installed to open any package's manual).

By the way, all citations, figure numbers, and section numbers are clickable links; if you want to see which objects are links, comment out `\hypersetup{hidelinks}` in `thesis.tex`. Links are a recent (within the last thirty years) innovation to TEX and are only availables with PDFs, so pdfLaTeX or some other PDF-first compiler is prefered. If you use XeLaTeX or LuaLaTeX you will likely need to change some of the packages loaded in `LA_Tech.cls` and in `thesis.tex`.

This document has been configured to use BibLATEX [7] to automatically typeset your bibliography. You will need to occasionally run the command `biber thesis.bcf` if you do not use a program such as `latexmk` to manage LATEX compilation. All you need to do is use `\cite{...}` with a given key from `bib.bib`, and everything will be taken care of. There are a variety of bibliography managers you might use rather than directly editing your bib file. Bibdesk comes with MacTEX, but most distributions don't include a manager. Consult `en.wikipedia.org/wiki/Comparison_of_reference_management_software` for a suitable application. I set the bibliography to use IEEE style, but it supports a variety of needs. You only need to change the options in `\usepackage[...]{biblatex}` in `thesis.tex` to suit most formatting options.

A more thorough introduction to contemporary LATEX can be found in the LATEX Wikibook [8] or at latex2e.org [9].

# CHAPTER 2

# TYPESETTING IN LaTeX

You can see how the various types of environments are created by looking at the corresponding source code.

## 2.1 Section Title

**Theorem 2.1.** *Let's say this is our first theorem.*

**Proof.** Type proofs as regular paragraphs. LaTeX will treat them as regular paragraphs and automatically format them consistently.

**Definition 2.2.** *Let's say this is our first definition.*

Here's how mathematics can be displayed. Note that a one-line "paragraph" before an itemized or enumerated list can look a bit funny, because the indentations don't match up.

- In-text math mode: $\int_0^1 x^2 dx = \frac{1}{3}$.
- Display math mode: To assure the proper spacing, the `$$...$$` environment must start on the line *immediately* after the preceding text.
$$\int_0^1 x^2 dx = \frac{1}{3}$$
- Display within text math mode: $\displaystyle\int_0^1 x^2 dx = \frac{1}{3}$

- And here is how we line up equalities. To assure the proper spacing, the `\begin{align*}` must be on the line *immediately* after the preceding text.

$$\begin{aligned}
\int_0^1 x^2 dx &= \left.\frac{1}{3}x^2\right|_0^1 \\
&= \frac{1}{3}
\end{aligned}$$

Note that delimiters like braces can be automatically adjusted to the right size by using the `\left` and `\right` commands. The above shows that the delimiters themselves need not match in shape and that a period gives an unprinted dummy delimiter.

Align also works when things get ugly:

$$\begin{aligned}
&\int_\Omega |D^\alpha f(x) - D^\alpha f_a(x)|^p \ d\lambda(x) \\
&= \int_\Omega \left| \int_\Omega D^\alpha f(x) g_a(x-z) \ d\lambda(z) - \int_\Omega D^\alpha f(z) g_a(x-z) \ d\lambda(z) \right|^p \ d\lambda(x) \\
&\leq \int_\Omega \left( \int_\Omega \left| D^\alpha f(x) - D^\alpha f(z) \right| g_a(x-z) \ d\lambda(z) \right)^p \ d\lambda(x) \\
&\leq \int_\Omega \left( \left( \int_\Omega \left( \left| D^\alpha f(x) - D^\alpha f(z) \right| \left( g_a(x-z) \right)^{\frac{1}{p}} \right)^p \ d\lambda(z) \right)^{\frac{1}{p}} \times \right. \\
&\qquad \left. \times \left( \int_\Omega \left( \left( g_a(x-z) \right)^{1-\frac{1}{p}} \right)^q \ d\lambda(z) \right)^{\frac{1}{q}} \right)^p \ d\lambda(x) \\
&= \int_\Omega \int_\Omega \left| D^\alpha f(x) - D^\alpha f(z) \right|^p g_a(x-z) \ d\lambda(z) \left( \int_\Omega g_a(x-z) \ d\lambda(z) \right)^{\frac{p}{q}} \ d\lambda(x) \\
&\leq \int_\Omega \int_{B_a(0)} \left| D^\alpha f(x) - D^\alpha f(x+y) \right|^p g_a(-y) \ d\lambda(y) \ d\lambda(x) \\
&= \int_{B_a(0)} \int_\Omega \left| D^\alpha f(x) - D^\alpha f(x+y) \right|^p \ d\lambda(x) g_a(y) \ d\lambda(y) \\
&< \int_{B_a(0)} \nu g_a(y) \ d\lambda(y) = \nu.
\end{aligned}$$

When you need an equation to be numbered, use the `equation` environment

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t}dt \tag{2.1}$$

Similarly the `align` environment will number each line. Adding `*` suppresses numbering with many commands.

Matrices are made similarly to the `align` environment. The `amsmath` package include several convenient matrix environments depending on what kind of brace of bracket you want.

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ab - cd \tag{2.2}$$

## 2.2 Next Section Title

**Definition 2.3.** *Let's say this is our second definition.*

**Theorem 2.4.** *Let's say this is our second theorem.*

The block at the end of a proof can be generated like this. □

## 2.3 Enumeration and Itemization

**Enumeration** is done like this

1. First entry.
2. Second entry.

**Bullets** are done like this

- First bullet.
- Second bullet.

### 2.4 Things to Explore

Go ahead and move the theorems, definitions and items around. After compiling twice, the references will be correct.

- The first theorem is Theorem 2.1.
- The second theorem is Theorem 2.4.
- The first definition is Definition 2.2.
- The second definition is Definition 2.3.
- The Fourier Transform is Equation 2.1
- The first enumeration entry is 1.
- The second enumeration entry is 2.
- An entry in the bibliography is [10].
- Another entry in the bibliography is [11].

*Note that when referring to a figure with* `\ref{}` *or to citations with* `\cite{}`*, it's often appropriate to use a non-breaking space (∼) so that your reference to Theorem 2.4 isn't split between lines.*

### 2.5 Bad Line and Page Breaks

Although LaTeX is designed to produce pages that look good, sometimes it produces a "widow" (first line of a paragraph alone at the end of a page) or an "orphan" (last line of a paragraph alone at the start of a new page). To force a page break, use the `\clearpage` command.

To force a right-justified line break,
use the `\linebreak` command. (But don't spread lines as above. You can see that it looks funny.)

### 2.5.1 Long Subsection Headings Long Subsection Headings Long Subsection Headings

Sometimes a section or subsection heading can be a bit long. To accommodate width requirements for headings, use hard carriage returns (\\) to insert line breaks in the heading. These hard carriage returns will also be used in the table of contents, but I was told that that is acceptable. Remember to set up the heading in inverted pyramid form (each line is narrower than the previous line).

# CHAPTER 3

# PLACEMENT OF FIGURES AND TABLES IN LaTeX

LaTeX treats figures and tables as floating objects, placing them where the compiler feels it is best [13]. If you place a *float* after the paragraph in which it is first referenced, then it will always be placed on the same page or later in the document. The `flafter` package restricts this placement so that a float always comes after its first call-out; you may activate it by uncommenting `\usepackage{flafter}` in `thesis.tex`.

The compiler does not always make the optimal decision when placing floats because it always chooses the first acceptable option. The placement of floats must be optimized manually when finalizing a manuscript by changing their position within the source file and applying the optional position argument `[htbp]`—here, top, bottom, or its own page. Changing the argument to `[t]` restricts the float to the top of a page and adding `!` relaxes some of the restrictions the compiler has for rejecting positions.

The space between text and floats for `[tb]` is reliably correct, but because of how doublespacing is implemented, an extra line of space is present below `[h]` figures. To correct this, Figure 3.1 has a negative `\vspace{...}` command below the caption.

### 3.1 Figures

When creating figures in LaTeX, it is important to realize that different types of figures are imported differently, leading to different margins below the figure. For diagrams, as in Figure 3.1, TeXCAD [14] is recommended since it creates native vector images, and text in it will match the rest of your project. The line in Figure 3.1 is at height $y = 0$ in TeXCAD. The distance between the figure and the caption is supposed to be a double space. Hence your lowest object should be near $y = 0$ in TeXCAD. A double space should separate the figure from paragraph text above or below it, which is done in Figure 3.1.

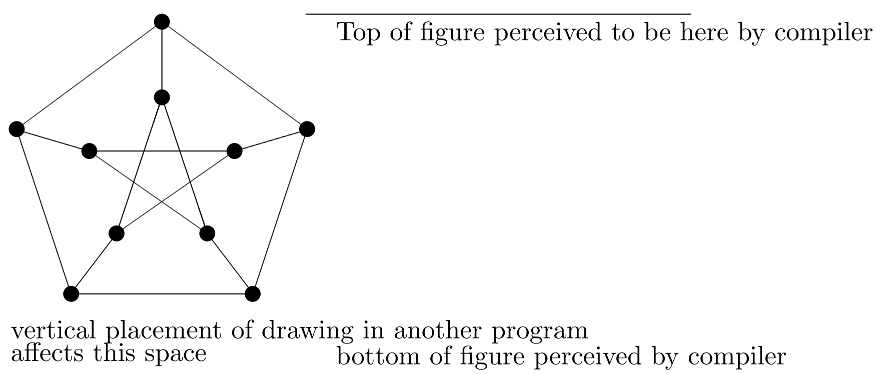


**Figure 3.1:** The line in the figure above is at height $y = 0$ in TeXCAD. Note that LaTeX automatically typesets captions single-spaced.

There will be more white space between the "bottom" of the figure and the caption if the lowest object is higher than $y = 0$. Figure 3.2 shows how items that are placed too high can distort the distance between the image and the caption. Additionally, notice how since it is positioned with `[t]` is does not need any additional spacing commands. Other packages for creating figures such as TikZ [15] automatically

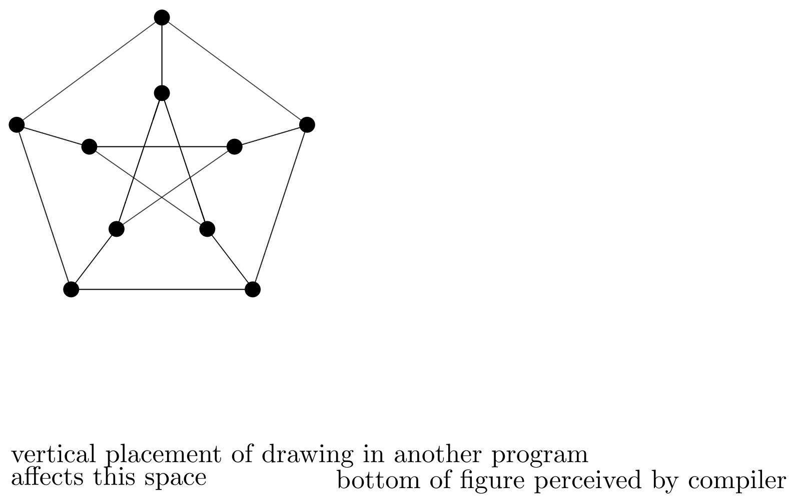


**Figure 3.2:** If your figure has some white space above the unseen actual border, like this one, only less extreme, please make a note on the printout that goes to the proofreader. Without the line on the bottom and the text, this figure would not be acceptable (too much white space). Disclaimer: This is an example of a **bad** figure with too much white space between content and bounding box.

create tight boxes around its drawings, but TikZ is also quite complicated and isn't for the light of heart.

Using `\includegraphics` from the `graphicx` package, a variety of image formats can be inserted with ease. In Figure 3.3, two JPEG files were imported. Their width was restricted to 40% of the width of the valid text area using the optional width argument. You might also want to restrict their height if the two images have different aspect ratios. You should only specify one of these so that the image is not warped. Note that imported pictures that have white space on their margins will look as if they are placed incorrectly. Such pictures should be cropped if possible. If they are not easily editable—such as a PDF—you should again use `\vspace` to correct its appearance.

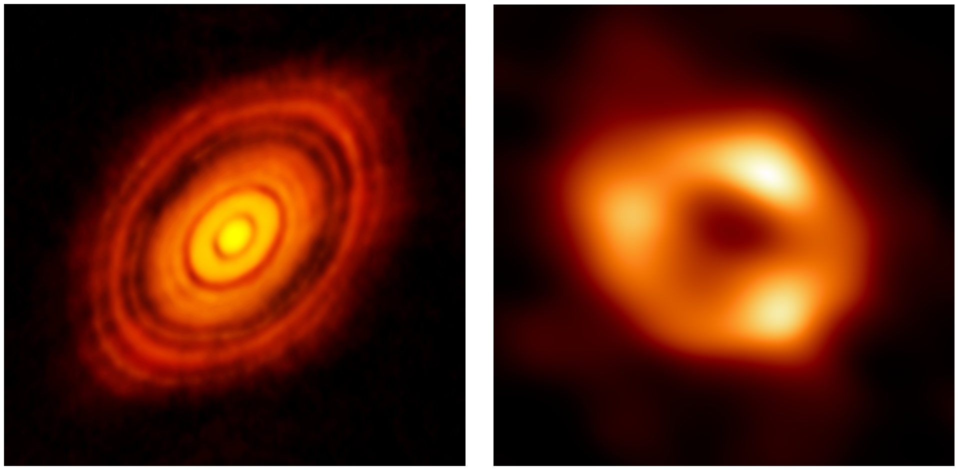

**Figure 3.3:** Two images captured using radio telescopes. The left is the HL Tauri protoplanetary disk, showing signs of planetary formation at only 100,000 years old [16]. The right is Saggitarius A$^*$, the black hole at the center of our galaxy [17].

## 3.2 Tables

Tables pretty much act like figures. Table 3.1 is an example of a small table. It was made using the built-in `tabular` environment. For tables, the title is to be placed above the table, which is done by putting the `\caption` command above the table. The alignment characters `lrcp` are required and control the alignment of each column. The optional vertical bar `|` controls the vertical lines separating columns.

For professional-looking tables, you'll want to minimize the number of lines used as done in Table 3.2. In addition, this table uses `\toprule`, `\midrule`, and

**Table 3.1:** A sample table. If there are no tables, comment out the `\listoftablestech` command in `thesis.tex`. *Note how the list of tables command spills too far to the right in the list of tables. This type of margin infraction must be corrected manually.*

| $p$ | $q$ | $p \Rightarrow q$ |
|---|---|---:|
| F | F | T |
| F | T | T |
| T | F | F |
| T | T | T |

\bottomrule from the package booktabs. These have improved vertical spacing over the built-in \hline and pleasantly split vertical lines. Again, since this table was placed using [h], an extra \vspace command is used to manually correct its spacing.

**Table 3.2:** Another sample table and another fun rule: Entries in the list of tables/figures should have at least 3 dots in the dotted line between entry and page number. Unless there is an unfortunate spill, which would need to be fixed anyway, LaTeX should do this automatically.

| $p$ | $q$ | $p \Rightarrow q$ |
|---|---|---|
| F | F | T |
| F | T | T |
| T | F | F |
| T | T | T |

More advanced tables might have parts than span multiple columns. LaTeX has a command to take care of this with \multicolumn, which is used in Table 3.3. The command \multirow works similarly and can be accessed with the package multirow. Additionally, this table has a single-line caption/title. Because of this, it has to be flush left with the table. There's no trivial way to do this (to my knowledge), so the table's width has been set usinng the tabularx environment. The capitalized versions of the normal alignment commands will make the rows expand to fill all excess space. This is done using the tabularx and array environments and is loaded automatically by LA_Tech.cls.

**Table 3.3:** Short name and multicolumn span

| Inputs | | | Output |
|---|---|---|---|
| $p$ | $q$ | $r$ | $p \oplus q \oplus r$ |
| T | F | Fdd | T |
| T | F | T | F |
| T | T | Fd | F |
| T | T | T | T |
| F | F | Fdd | F |
| F | F | T | T |
| F | T | F | T |
| F | T | Tdddd | F |

# CHAPTER 4

# CONCLUSIONS

Put your conclusions here.

The quick brown fox jumped over the lazy dog. (I'm so tired of animal acts.) The quick brown fox jumped over the lazy dog. (I'm so tired of animal acts.) The quick brown fox jumped over the lazy dog. (I'm so tired of animal acts.) The quick brown fox jumped over the lazy dog. (I'm so tired of animal acts.) The quick brown fox jumped over the lazy dog. (I'm so tired of animal acts.) The quick brown fox jumped over the lazy dog. (I'm so tired of animal acts.) The quick brown fox jumped over the lazy dog. (I'm so tired of animal acts.) The quick brown fox jumped over the lazy dog. (I'm so tired of animal acts.) The quick brown fox jumped over the lazy dog. (I'm so tired of animal acts.) The quick brown fox jumped over the lazy dog. (I'm so tired of animal acts.)

The quick brown fox jumped over the lazy dog. (I'm so tired of animal acts.) The quick brown fox jumped over the lazy dog. (I'm so tired of animal acts.) The quick brown fox jumped over the lazy dog. (I'm so tired of animal acts.) The quick brown fox jumped over the lazy dog. (I'm so tired of animal acts.) The quick brown fox jumped over the lazy dog. (I'm so tired of animal acts.) The quick brown fox jumped over the lazy dog. (I'm so tired of animal acts.) The quick brown fox jumped over the lazy dog. (I'm so tired of animal acts.) The quick brown fox jumped over the lazy dog. (I'm so tired of animal acts.) The quick brown fox jumped over the lazy dog. (I'm so tired of animal acts.) The quick brown fox jumped over the lazy dog. (I'm so tired of animal acts.)

The quick brown fox jumped over the lazy dog. (I'm so tired of animal acts.) The quick brown fox jumped over the lazy dog. (I'm so tired of animal acts.) The quick brown fox jumped over the lazy dog. (I'm so tired of animal acts.) The quick brown fox jumped over the lazy dog. (I'm so tired of animal acts.) The quick brown fox jumped over the lazy dog. (I'm so tired of animal acts.) The quick brown fox jumped over the lazy dog. (I'm so tired of animal acts.) The quick brown fox jumped over the lazy dog. (I'm so tired of animal acts.) The quick brown fox jumped over the lazy dog. (I'm so tired of animal acts.) The quick brown fox jumped over the lazy dog. (I'm so tired of animal acts.) The quick brown fox jumped over the lazy dog. (I'm so tired of animal acts.)

# APPENDIX A

# APPENDICES (IF DESIRED)

**Put your appendices here. All codes stay the same, except that the first page of an appendix should only have the headings, so follow `\chapter` with a `\clearpage`. Code in `thesis.tex` take care of naming a `\chapter` an "APPENDIX."**

From Louisiana Tech's "GUIDELINES FOR THE PREPARATION AND SUBMISSION OF YOUR THESIS OR DISSERTATION" (see [18])

- Appendices are optional. They must contain extra, relevant material such as questionnaires, surveys, tables, figures, computer data, and letters of permission to reprint copyrighted material. These optional appendices must be listed in the Table of Contents, conforming to the format used there. They must also be formatted in the document in such a way that they are consistent with the other main divisions.
- Appendices must be cited in the Body of the document.
- All material in appendixes must be numbered consecutively, within the required margins, and on the same paper used throughout the document.
- All appendices must have a title page. The title page of each Appendix must have the Arabic page number centered between the left and right margins between 1/2 inch and 1 inch from the bottom edge of the page. Place the Arabic page number at the top right of subsequent pages of each Appendix.
- The Title page must have the word APPENDIX typed in all capital letters 2 inches from the top of the page followed by an informative title in all capital letters. If there is more than one appendix, then label the first title page as

APPENDIX A, the second APPENDIX B, and so on, providing an informative title for each appendix.

# BIBLIOGRAPHY

[1] [Online]. Available: `http://miktex.org/`

[2] [Online]. Available: `http://winedt.com/`

[3] [Online]. Available: `http://www.texniccenter.org/`

[4] [Online]. Available: `https://tug.org/mactex/mactex-download.html`

[5] [Online]. Available: `https://www.tug.org/texlive/`

[6] [Online]. Available: `https://www.overleaf.com`

[7] [Online]. Available: `https://ctan.org/pkg/biblatex`

[8] [Online]. Available: `https://en.wikibooks.org/wiki/LaTeX`

[9] *Latex2e: An unofficial reference manual*. [Online]. Available: `latex2e.org`

[10] A. A, *An entry in the bibliography*, For bibliographies single space the entries and put a double space between entries.

[11] A. B, *Another entry in the bibliography*.

[12] A. C, *Third entry in the bibliography*, Caution: Use the formatting style that is common in your discipline. *None of the entries in this file should be considered a sample entry.*

[13] F. Mittelbach, "How to influence the position of float environments like figure and table in LaTeX," *TUGboat*, vol. 25, no. 3, 2014. [Online]. Available: `https://www.latex-project.org/publications/2014-FMi-TUB-tb111mitt-float-placement.pdf`

[14] [Online]. Available: `http://texcad.sourceforge.io/`

[15] *Pgf – create postscript and pdf graphics in TEX*. [Online]. Available: `https://www.ctan.org/pkg/pgf`

[16] ALMA Partnership et al., "The 2014 ALMA Long Baseline Campaign: First Results from High Angular Resolution Observations toward the HL Tau Region," *APJL*, vol. 808, no. 1, Jul. 2015, Art. no. L3. DOI: `10.1088/2041-8205/808/1/L3` arXiv: `1503.02649 [astro-ph.SR]`.

[17] *First image of our black hole*. [Online]. Available: https://www.eso.org/public/images/eso2208-eht-mwa/

[18] *Guidelines for the preparation and submission of your thesis or dissertation*, Louisiana Tech Univervsity. [Online]. Available: http://www.latech.edu/graduate_school/thesis_dissertations/index.shtml

# VITA (IF REQUIRED)

- The Vita is a one-page biographical sketch of the author written in paragraph form and in 3rd person. It is the last item of the document and must appear in the Table of Contents.
- The heading must have identical font, value, size, and position/location in the page as other major headings.